UNIVERSITY OF GREENWICH

Faculty of Business | School of Business Operations and Strategy

Dissertation Research Proposal

Title

Predicting Customer Churn in Motor Insurance: A Machine Learning Approach to Customer Retention

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Introduction

One of the biggest business problems the insurance industry is facing is customer churn, or the voluntary cancellation of a car insurance policy due to non-renewal or lapse. When price competition dominates the market and the acquisition of existing policyholders is expensive, it is far more cost effective to keep customers. However, many insurers are still using experience-driven, rather than data-driven, strategies for retention.

This research proposal presents a Traditional Research Project where machine learning classification methods will be used for a customer churn prediction in the Motor Vehicle Insurance domain. A set of 105,555 policy records will be used, from the Inter-university Consortium for Political and Social Research (ICPSR), a trusted academic open-data repository, with 30 variables covering demographics, policy features, premium and payment patterns, claims history, and vehicle attributes.

Project Type:

It is a Traditional Research Project which will analysis a large secondary dataset to address defined research questions using quantitative machine learning methods.

Research Questions:

RQ1: Which machine learning classification model best predicts motor insurance customer churn under real-world class imbalance conditions?

RQ2: Which policy, customer, and vehicle features most strongly predict churn, and how can these drive retention strategy?

The goals of the study include exploratory data analysis, feature engineering, model building and comparison, model evaluation, and business recommendations based on the model's predictions. The project is based on the CRISP-DM (Cross-Industry Standard Process for Data Mining) structure, which is a structured analytical method that aligns with the process from problem definition to strategy recommendation.

This study has made a significant contribution to the applied machine learning literature in the field of insurance analytics by systematically comparing the performance of popular classification algorithms on motor insurance lapse data, in a reproducible way. The value of the practical aspect is its ability to provide insurance carriers with a valid, explainable churn prediction model to help them implement proactive and targeted retention actions.

Literature Review

2.1 Churn Prediction in Financial Services

The prediction of customer churn has received significant research interest in many industries, such as telecommunications, banking, and retail. In all these areas, it has been found that machine learning models (especially ensemble models) achieve higher predictive accuracy than traditional statistical models (Ngai, Xiu and Chau, 2009). The use of these techniques in insurance, however, and in motor insurance in particular is still relatively underdeveloped in the published literature and it is this area of insurance that the present project aims to fill.

The first studies on predicting churn used logistic regression as the main technique because it was easy to interpret and because it provides output in a probability format (Verbeke et al., 2012). Although logistic regression is a well-established and widely used baseline model for binary classification problems, tree-based and ensemble models have consistently outperformed logistic regression in predictive accuracy with complex and non-linear insurance data (Loisel and Piette, 2018). The advantage of decision trees is that they can be used to create human interpretable classification rules that can be used directly in business decisions while random forests, being an ensemble of decision trees, produce improved generalisation, variance reduction, and feature importance estimation (Breiman, 2001).

2.2 Class Imbalance in Insurance Churn Data

One of the most important methodological issues in churn prediction is that of class imbalance, whereby the churned class is a minority of the observations, and the classic classifiers suffer from systematic bias towards the majority class (Burez and Van den Poel, 2009). About 79.6% of the observations in this dataset were of active policyholders who did not have any customer churn history, while 18.9% were of churned customers and the remainder were of multi-lapsed observations. This imbalance is similar to averages drawn from the literature among all insurance portfolios and is a common analytical challenge.

Class imbalance is a topic that has been discussed in the methodological literature, and study shows that it is better to tackle the imbalance before model training, rather than tolerance biased accuracy as the main metric for model performance (Chawla et al., 2002). A practically and computation efficient method for this is cost-sensitive learning, which is achieved by using the class_weight argument in scikit-learn, giving the minority churned class more importance when training the model without having to expand the minority class using augmentation techniques.

2.3 Theoretical Framework and Research Gap

This project is based on the theoretical concept of Predictive AI, which is a model of the CRISP-DM process that uses historical customer behaviours to predict future actions. CRISP-DM offers an iterative, popular approach to applied analytics projects in insurance and financial service settings (Chapman et al., 2000). The six phases of its development are: business understanding, data understanding, data preparation, modelling, evaluation, and deployment, and are aligned with the analytical process used in this dissertation.
Customer tenure, premium level, claims frequency and payment behaviour have generally been found to be important determinants of customer lapse in insurance-specific contexts (Guillen et al., 2012). All these variables and more are included in the present dataset, including vehicle attributes and distribution channel data, which allows for a well-specified predictive model. There has been little work directly comparing standard classification algorithms, such as KNN, SVM, tree-based models and so on, for motor insurance data, in the presence of class imbalance and in the explicit context of retention strategy. This project directly fills that gap. In all churn prediction studies reported (He et al., 2014), feature engineering, which involves creating analytically meaningful variables from raw data, has consistently been found to be a key to good model performance. The calculated attributes (derived columns) in this dataset are likely to be important in predicting the claim-to-premium ratio beyond the raw data columns.

Methodology

This project is a quantitative, deductive research design, which involves testing and refining theoretically driven expectations on motor insurance churn predictors based on the data collected. The methodology is based on the six phases of CRISP-DM.

Phase 1 : Business Understanding:

The business problem is translated to a binary classification problem: for a collection of customer, policy, and vehicle attributes at a point in time (of a policy year), classify whether the customer will allow their policy to lapse during that time.

Phase 2 : Data Understanding:

Exploratory Data Analysis will be performed using pandas and seaborn to investigate the distributions of the variables, the absence of variables and bivariate association between the variables (predictors) and the target variable (churn). This data set has 105,555 samples and 30 attributes. Some of the key variables are Lapse (target), Seniority (customer tenure), Premium, Cost_claims_year, N_claims_year, R_Claims_history, Payment (channel), Distribution_channel, Type_risk, Area, Second driver, and vehicle-level attributes (Year_matriculation, Power, Cylinder_capacity, Value_vehicle, Type_fuel). Within Type_risk group, mode imputation is used for Type_fuel, as there are 1,764 (1.7%) data missing from this feature; it will also be removed from the feature set unless imputation is warranted by EDA. Length is not imputed as there are 10,329 (9.8%) missing values and will be excluded from the feature set unless imputation is justified by EDA. There are 70,408 records that do not contain a Date_lapse field, as expected for non-lapsed policies, which will not be used as a predictor.

Phase 3 : Data Preparation:

The Lapse variable will be binarized, 0 = retained; 1 = churned (Lapse ≥ 1). Continuous date fields (Date_birth, Date_start_contract, Date_driving_licence) will be transformed into derived: customer age, policy tenure (years) and driving experience (years). Categorical variables, such as Distribution_channel, Type_risk, Area, Type_fuel and Payment, will be encoded using one-hot encoding. Continuous features will be standardised before modelling with distance-based and margin-based algorithms (KNN, SVC) and not standardised for tree-based algorithms. All preprocessing will be done using a scikit-learn Pipeline to avoid data leakage. The dataset will be divided into 70% training and 30% testing sets, and the Stratified K-Fold cross-validation (k=5) method will be used to ensure the even distribution of class proportions across the folds.

Phase 4 : Modelling:

To prepare the data, five classification algorithms are to be trained: Logistic Regression, Decision Tree, Random Forest, KNN, and SVC. To handle the class imbalance, all models will be tuned with class_weight='balanced'. After that, hyperparameter tuning using the grid search cross validation technique will be employed on the best performing model(s) based on the F1-Score and ROC-AUC score on the validation folds.

Phase 5 : Evaluation:

Stratified k-fold cross validation (k=5) will be used for all models to tune their parameters using GridSearchCV or RandomizedSearchCV. The data set will be divided into a training set (70%), a validation set (15%) and a held-out test set (15%) all of which will be split so that the proportions of classes are preserved. The importance of features (Decision Tree, Random Forest) and magnitude of the coefficients (Logistic Regression) will be used to determine the most important features to predict churn.

Phase 6 : Business Recommendations:

Results will be converted to practical recommendations for retention strategies. High churn-risk customers will be segmented based on their main risk profile (high claims, low tenure, low value vehicle, non-direct payment channel) to ensure personalised interventions such as cross-selling, reminder campaigns for renewals, and premium discounts.

Ethical Considerations:

The data comes from an institutional repository, ICPSR (Open ICPSR), which meets the standards for data governance and privacy. There is no personally identifiable information (PII) contained in the data set, instead, all customer records are represented by an anonymised integer ID. The data set will not be disseminated and will be used solely for academic research. The data, the compiled Python notebook, and README describing the data source will be provided in the GitHub repository, in accordance with module requirements. The use of any AI tool in the research process will be acknowledged as per the University of Greenwich AI Usage Guidance (2024).

Limitations:

They are a secondary data set, which means that the data does not necessarily represent the operational context of any individual insurer, and that generalisability of model results to real-world deployment environments cannot be guaranteed without additional validation. Lack of contextual variables for the macroeconomic level (market premium indices, competitor pricing) can limit the explanatory power of the model. Explicitly these limitations will be discussed in the project report.

A link to the GitHub repository with the dataset, Python notebook and README will be included in the Methodology chapter of the final project submission.

Project Management

The project will run over a 16-week period from the submission of the project proposal to the submission of the project, as outlined in table 1 below.

Weeks	Phase	Activities & Milestones
1–2	Business Understanding	Finalise research questions; set up GitHub repository; review supervisor feedback on proposal.
3–4	Data Understanding (EDA)	Complete exploratory data analysis; produce visualisations; document dataset characteristics.
5–6	Data Preparation	Feature engineering; missing data handling; encoding; train/test split; pipeline construction.
7–9	Modelling	Train all five classifiers; apply class weighting; cross-validation; baseline performance comparison.
10–11	Evaluation & Tuning	Hyperparameter tuning; test set evaluation; metric reporting; ROC curve generation.
12–13	Literature Review	Critical literature synthesis; research gap articulation; referencing (Harvard style).
14–15	Discussion & Recommendations	Interpret findings; develop retention strategy framework; write Discussion and Conclusion chapters.
16	Submission	Final proofreading; GitHub finalisation; Turnitin submission by deadline.

Table 1: Project timeline and milestone plan.

Resource requirements are minimal:

Python 3.x with scikit-learn, pandas, NumPy, matplotlib, and seaborn installed; a GitHub account for repository management; and access to the University of Greenwich library for academic literature. No primary data collection is required, and no software licences beyond freely available open-source tools will be needed.

Risk mitigation strategies include:

- weekly supervisor meetings to monitor progress and address methodological uncertainties early.
- version-controlled code in GitHub to prevent data loss.
- a modular notebook structure to allow individual components to be revised independently and
- a contingency allocation of two weeks within the timeline to absorb delays.

Expected Outcomes

The project is expected to produce the following deliverables and contributions:

- A comparative study of five machine learning classifiers, comparing the performance of the classifiers with respect to the standard classification measurements and paying particular attention to class imbalance, to determine which model is best suited to classify motor insurance customers at risk of churn.
- An analysis of the feature importance with the customer and the vehicle and policy attributes most strongly correlated to policy lapse, giving actionable insight to the retention teams.
- A set of recommendations that address customer retention based on the data, segmented at-risk customers based on dominant churn risk profile, and suggesting targeted interventions scaled to the estimated customer lifetime value.
- A complete Python notebook and GitHub repository with professional-quality code and reproducible analysis.

Success will be judged upon the following: ROC-AUC value of the best model of at least 0.70, the F1-Score of the winning model is greater than the F1-Score of the majority-classifier as a baseline, and the model identifies at least three statistically significant predictors of churn that support the existing literature. After finishing the results, additional iterations of feature engineering and hyperparameter optimisation will be conducted if the model performance is below these thresholds.

The overall value of this project is in providing a proof of concept on the use of predictive AI in a business analytics structured framework to provide commercially actionable intelligence. The project puts machine learning outputs in the context of a retention strategy, connecting modelling with the managerial decision-making process, which is increasingly valued by financial services data science and business intelligence professionals. Valued across data science and business intelligence roles in the financial services sector.

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Declaration of AI:

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic – NO
- To create an outline of the topic – YES
- To explain concepts – NO
- To support my use of language – YES